01

INCINERATOR

IN HOUSE HOLD

5 Weeks 1

How might we reduce harmful emissions from incinerators and create a cleaner, more efficient waste-processing system?

Context

Household waste incineration can provide a convenient way of managing waste, but inefficient combustion can result in black fumes and harmful emissions. The project explores how emission control can be made more suitable for small-scale incineration, where conventional industrial solutions may be too expensive or complex.

The problem of waste burning at a household scale

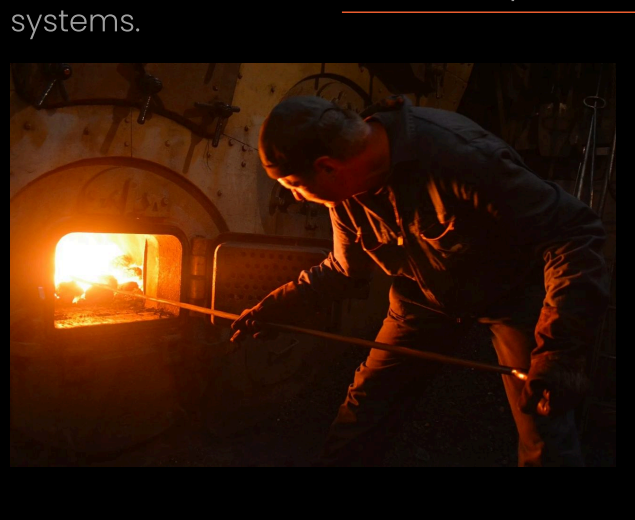
Why This Project

The project began with the observation of black fumes being released from an incinerator in Malappuram , Kerala. This raised a key question:

How might we reduce harmful emissions from small-scale incinerators without relying on complex and expensive industrial filtration systems?

Malappuram , Kerala.

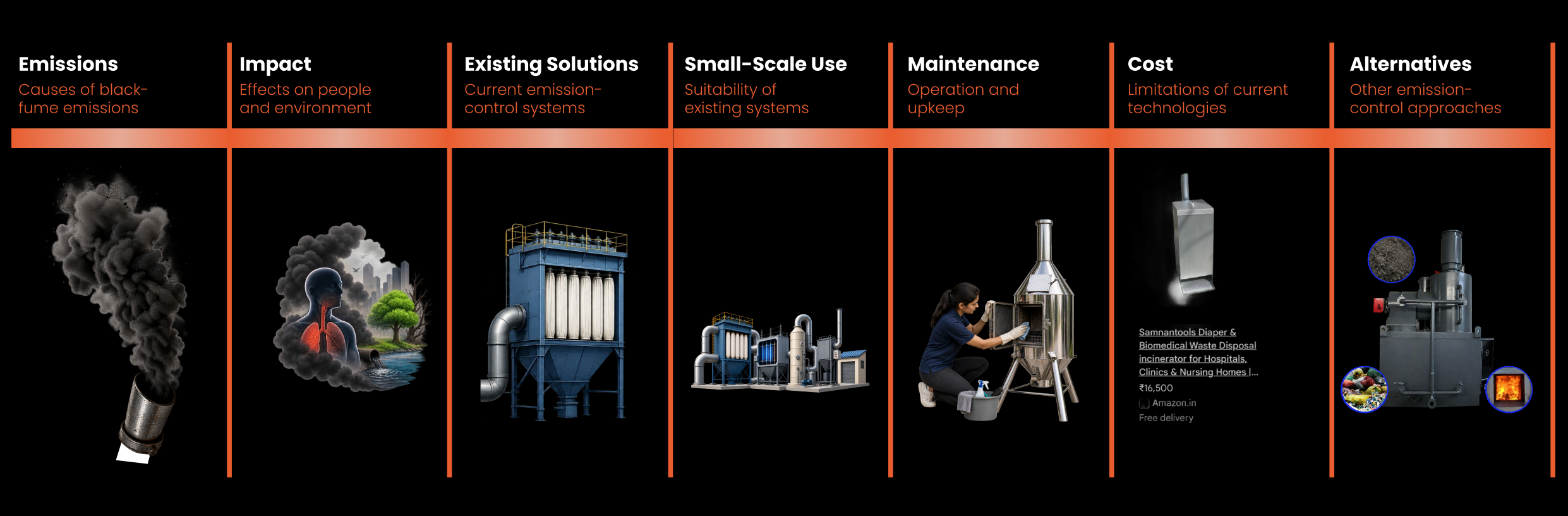
The focus shifted towards developing a solution that could be compact, feasible and adaptable to smaller incineration systems.



Initial

Research

- Conducted secondary research on **7 existing small-scale and household incinerator products** available in the market.
- Studied products across different categories, including **household waste, sanitary waste and small institutional incinerators.**
- Compared the products based on key features such as **capacity, size, materials, energy source, operation, insulation, exhaust system, safety, maintenance and portability.**
- Analysed the products using a **competitive benchmarking matrix** to identify their strengths, limitations and areas of improvement.
- The comparison showed that existing products generally focus on **waste capacity and combustion efficiency,** while fewer products focus on **compactness, intuitive interaction and household-friendly aesthetics.**
- The research helped identify an opportunity for a **compact, safe and easy-to-use household-oriented solution.**



Emissions	Impact	Existing Solutions	Small-Scale Use	Maintenance	Cost	Alternatives
Causes of black-fume emissions	Effects on people and environment	Current emission-control systems	Suitability of existing systems	Operation and upkeep	Limitations of current technologies	Other emission-control approaches

Samnantools Diaper & Biomedical Waste Disposal incinerator for Hospitals, Clinics & Nursing Homes I...
₹16,500
Amazon.in
Free delivery

Existing solutions such as baghouse filters, ESPs and scrubbers were studied. While effective in specific applications, their cost, maintenance requirements and limitations for gaseous pollutants revealed an opportunity for a simpler small-scale approach.

THE PROBLEM

Black Fumes from Incinerators

Incinerators play an important role in waste management, particularly for hazardous, medical and municipal waste. However, inefficient combustion and inadequate filtration can result in the release of black fumes and contaminated air.

These emissions can create serious risks for both human health and the environment.

The Challenge

How might we reduce harmful emissions from incinerators and create a cleaner, more efficient waste-processing system?

TOUCH POINTS

02 — Complexity

Maintenance and operation can make conventional systems difficult to manage at a smaller scale.

03 — Adaptability

A smaller, modular system could potentially be integrated with existing incinerators instead of replacing the entire system.

01 — Cost

Industrial emission-control systems can require significant investment.

INSIGHTS

01 High Cost

Advanced emission control can be costly and complex.

02 Incomplete Combustion

Produces black smoke, soot and harmful emissions.

03 Filtration Limitations

Fine and gaseous pollutants can be difficult to capture.

04 High Maintenance

Filters need regular cleaning and replacement.

05 Large-Scale Systems

Many solutions are designed for larger facilities.

06 Integrated Approach

Better combustion and filtration are needed together.

PRIMARY RESEARCH

- Site observations and interviews with facility staff to understand current incinerator operations.
- Questionnaire / survey conducted among 35 users along with people who are effected.
- Observation of users using incinerators.

80% Users feel that manual methods do not extract juice efficiently.

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60% Feel that products with longer usability can reduce unnecessary replacement.

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Key Insight

Users want a juicer that saves time and effort, provides maximum juice extraction and is easy to clean and operate.

USER PERSONA 01

NAME: Rajesh
AGE: 45
ROLE: Incinerator Operator
WORK ENVIRONMENT: Waste-management facility

Key Quote
"The system should control emissions without making operation and maintenance more complicated."

Goals
- Operate the incinerator efficiently
- Maintain proper combustion
- Control harmful emissions
- Avoid regulatory violations
- Minimize downtime and maintenance

Needs
- Reliable emission-control equipment
- Easy maintenance
- Simple monitoring
- Affordable operation
- Clear indication of system performance

Pain Points
- Complex maintenance
- High equipment costs
- Frequent filter replacement
- Difficulty monitoring emissions continuously
- Existing systems can be expensive and unsuitable for smaller facilities

USER JOURNEY

STAGE	COLLECT WASTE	SORT & PREPARE	LOAD INCINERATOR	BURN WASTE	MONITOR EMISSIONS
ACTION	Checks and prepares waste for burning	Loads waste into the chamber	Starts and operates the incinerator	Observes smoke and system performance	Checks filters and equipment
THINKS	"Is this waste suitable for burning?"	"Is the chamber loaded properly?"	"Is the combustion working properly?"	"Why is the smoke becoming black?"	"Does the filter need cleaning?"
FEELS	Focused	Careful	Alert	Concerned	Frustrated
PAIN POINT	Difficult waste handling	Manual loading	Incomplete combustion can create black fumes	Limited continuous monitoring	Maintenance can be complex
NEED	Simple preparation	Safe loading system	Efficient combustion	Clear emission indication	Easy filter access and maintenance

EMAPTHY MAPPING

SAYS
"I need to keep the incinerator running."
"The smoke is becoming black."
"Maintenance takes time."

DOES
Collects and handles waste
Sorts and prepares waste
Loads waste into the incinerator

THINKS
"Is the waste burning properly?"
"What is causing the black fumes?"
"Can I maintain this system more easily?"

FEELS
Responsible
Focused
Careful

PAIN
- Black fumes during burning
- Difficult to identify emission problems
- Limited continuous monitoring

GAIN
- Cleaner exhaust emissions
- Easy-to-understand monitoring
- Efficient waste combustion

USER PERSONA 02

NAME: Anitha
AGE: 38
ROLE: Local Resident / Shop Owner
WORK ENVIRONMENT: Residential area near waste-management facility

Key Quote
"I want clean air around my home and a safe environment for my family."

Goals
- Live in a clean and safe environment
- Reduce exposure to harmful emissions
- Protect family health
- Understand the source of pollution
- Ensure proper emission control

Needs
- Cleaner air around the facility
- Reduced black-fume emissions
- Reliable emission monitoring
- Clear information about air quality
- Effective pollution control

Pain Points
- Frequent black fumes
- Poor local air quality
- Uncertainty about harmful emissions
- Lack of visible emission monitoring
- Concern about long-term health effects

USER JOURNEY

STAGE	LIVES NEAR FACILITY	SEES BLACK FUMES	EXPERIENCES POLLUTION	SEEKS INFORMATION	MONITOR EMISSIONS
ACTION	Goes about her daily routine	Notices dark smoke	Experiences smoke and unpleasant air	Talks to neighbours / looks for information	Expects authorities to address the issue
THINKS	"Is the facility operating safely?"	"Why is the smoke black?"	"Is this harmful to my family?"	"What is being released into the air?"	"Will the pollution be controlled?"
FEELS	Comfortable	Concerned	Worried	Frustrated	Hopeful
PAIN POINT	Difficult waste handling	Black-fume emissions	Poor air quality	Lack of clear information	Lack of visible control
NEED	Safe surroundings	Cleaner emissions	Healthy air	Reliable information	Effective emission control

EMAPTHY MAPPING

SAYS
"Why is the smoke becoming black?"
"The air quality is getting worse."
"Is this harmful to my family?"

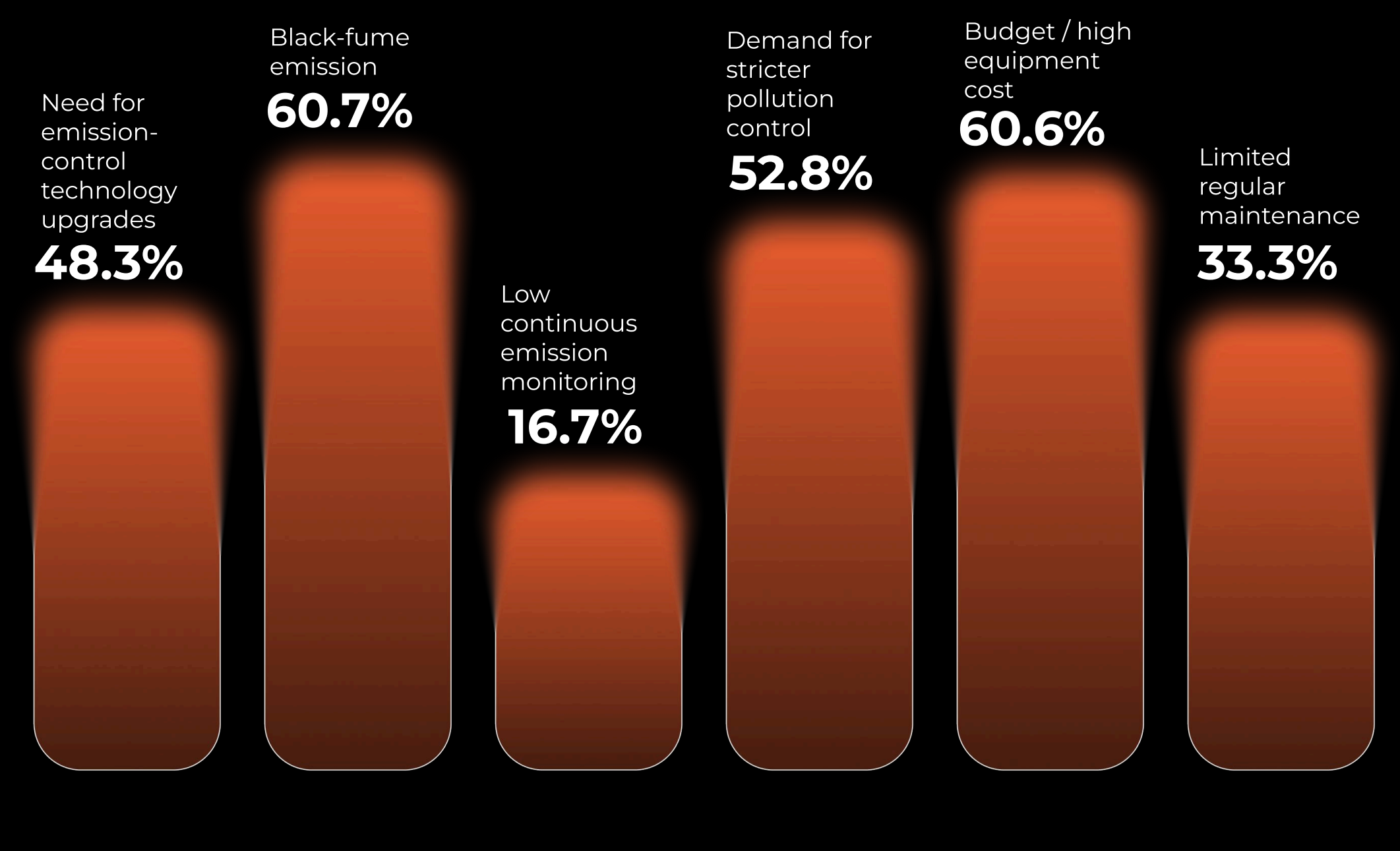
DOES
Observes the smoke
Talks to neighbours
Looks for information about pollution

THINKS
Observes the smoke
Talks to neighbours
Looks for information about pollution

FEELS
Concerned
Worried
Unsafe

PAIN
- Observes the smoke
- Talks to neighbours
- Looks for information about pollution

GAIN
- Cleaner air
- Reduced black fumes
- Reliable emission monitoring



Problem statement

The lack of a cost-effective and easily maintainable emission-control system for incinerators results in uncontrolled black-fume and harmful gas emissions.

Design Direction

01 Modular Filtration

02 Easy Maintenance

03 Efficient Emission Control

Develop a modular catalytic filtration unit that can be integrated into existing incinerators for improved emission control.

Design the system for simple access, inspection, and replacement of filtration components to reduce maintenance effort.

Use multi-stage catalytic treatment to reduce black fumes and harmful gases, resulting in cleaner exhaust emissions.